Higher-order transformations of bidirectional quantum processes

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1 Introduction

An important class of quantum devices is bidirectional, meaning that the roles of input and output ports can be interchanged, for instance, optical elements such as half- and quarter-wave plates, which can be traversed in opposite directions and generally induce different polarization rotations. Mathematically, these devices correspond to bistochastic quantum channels, i.e., completely positive maps that are both trace-preserving and identity-preserving. Such channels are central in quantum thermodynamics, majorization theory, and recent foundational investigations of time symmetry. Beyond using a bidirectional device in one direction or the other, recent work has shown that quantum mechanics allows for a more radical possibility: operations in which the input-output direction itself becomes indefinite, for example when a system traverses a device in a coherent superposition of opposite directions, a primitive known as the quantum time flip [5]. This phenomenon, termed input-output indefiniteness, has applications in channel discrimination, communication, metrology, thermodynamics, and the study of quantum causality, and has been experimentally demonstrated in photonic platforms. While the basic framework for operations with indefinite input-output direction was previously developed for transformations mapping several bistochastic channels into a single output channel, a complete characterization of general higher-order transformations, allowing arbitrary numbers of input and output channels, remained open.

In this contribution, we fill this gap by providing a full classification of admissible transformations with indefinite input-output direction [1]. We introduce an infinite hierarchy of higher-order maps built on bistochastic channels, identify the allowed structures at each level, and uncover processes in which channels are combined in definite causal order while each individual channel is used with indefinite direction. For this class, we prove a realization theorem showing that they can be decomposed into causally ordered sequences of transformations. Our approach builds on the theory of quantum supermaps and higher-order quantum transformations [2], extending it to the restricted setting of bistochastic channels, which can be interpreted as dynamics in a time-symmetric variant of quantum theory. The resulting hierarchy characterizes the largest set of logically consistent processes compatible with this time-symmetric framework, thereby deepening the connection between higher-order quantum operations, causal structure, and time symmetry.

2 Framework

In order to construct a theory with arbitrary higher-order quantum maps, we use a formal language of type system [2]. It starts with the fundamental block of elementary types labeling finite-dimensional quantum systems with a Roman letter (A, B, etc.). By combining them with brackets and arrows, one can construct new types recursively: if $\mathtt{x}$ and $\mathtt{y}$ are types, then $(\mathtt{x} \to \mathtt{y})$ is a type as well. For example, a quantum channel corresponding to a quantum device with input port A and output port B belongs to type $(A \to B)$, while a higher-order transformation mapping quantum channels to quantum channels belongs to type $((A \to B) \to (C \to D))$. Finally, if a quantum device is bidirectional, so that its ports A and B can be exchanged, the corresponding systems are equipped with a circumflex accent, e.g., $(\hat{A} \to \hat{B})$ (see Fig. 1). In such a way, a whole hierarchy of types of higher-order transformations with bidirectional quantum devices can be generated.

Submitted to:
LIQCS 2026

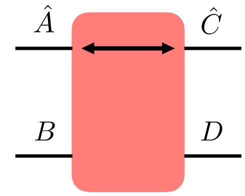

Figure 1: Partially bistochastic channel.

Since any type is an abstract string of symbols, it is necessary to provide a bridge between types and physical transformations described by linear maps on Hilbert spaces. This is done by the notion of event, which is defined in a recursive manner similarly to the type system. Events of an elementary type A are bounded operators on the corresponding Hilbert spaces $\mathscr{H}_A$, so that they form a set $T_{\mathbb{R}}(A) = \mathscr{L}(\mathscr{H}_A)$. In turn, if $\mathtt{x}$ and $\mathtt{y}$ are types, then events of type $(\mathtt{x} \to \mathtt{y})$ are Choi operators of maps sending $T_{\mathbb{R}}(\mathtt{x})$ to $T_{\mathbb{R}}(\mathtt{y})$, which form a set $T_{\mathbb{R}}(\mathtt{x} \to \mathtt{y}) = \mathscr{L}(\mathscr{H}_{\mathtt{x}} \otimes \mathscr{H}_{\mathtt{y}})$, where $\mathscr{H}_{\mathtt{x}/\mathtt{y}} = \bigotimes_i \mathscr{H}_i$ with i going over the set of elementary types occurring in the type $\mathtt{x}$ or $\mathtt{y}$, respectively. Obviously, since not every bounded operator on $\mathscr{L}(\mathscr{H})$ is physical (e.g., representing a physical state or map), it is necessary to restrict the introduced set of events. For this purpose, for a given type $\mathtt{x}$, we define the subset $T(\mathtt{x}) \subset T_{\mathbb{R}}(\mathtt{x})$ of admissible events. It consists of Choi operators such that the corresponding higher-order map satisfies certain minimal admissibility requirements that ensure its compatibility with the probabilistic structure of quantum theory. These requirements are a recursive generalization of the admissibility conditions in Kraus' axiomatic approach to quantum operations: i) linearity, ii) complete positivity, and iii) subnormalization. Linearity is straightforwardly extended, while complete positivity entails that an admissible map of type $\mathtt{x} \to \mathtt{y}$ should transform admissible maps of type $\mathtt{x}$ into admissible maps of type $\mathtt{y}$, even in the presence of correlations with another system E. This notion necessitates defining an extension of a type $\mathtt{x}$ with an elementary type E. The subnormalization condition guarantees that the generated probabilities are upper-bounded by 1. Finally, we define the central object of the framework, the subset $T_1(\mathtt{x}) \subset T(\mathtt{x})$ of deterministic admissible events of type $\mathtt{x}$ under a stronger condition of normalization which corresponds to preservation of the generated probabilities.

In order to characterize arbitrary physical higher-order transformations, we prove a characterization theorem for admissible higher-order quantum maps with quantum bidirectional devices. This result is transparent in terms of the Choi isomorphism: the admissible events $R \in T_1(\mathtt{x})$ of arbitrary type $\mathtt{x}$ are positive operators $R = \lambda_{\mathtt{x}} I_{\mathtt{x}} + X_{\mathtt{x}}$, with each type translating into a set of linear constraints on $\lambda_{\mathtt{x}}$ and subspaces $\Delta_{\mathtt{x}} \ni X_{\mathtt{x}}$ of space of traceless operators on $\mathscr{H}_{\mathtt{x}}$.

3 Networks of higher-order maps and bistochastic generalization of quantum combs and process matrices

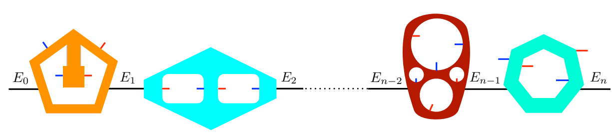

Figure 2: A network of higher-order maps.

Indefinite input-output direction captures the idea that a physical device may not possess a predetermined orientation between its input and output systems: the same operation can be used in either direction. While a single bistochastic device already allows for such local bidirectionality, one can also consider scenarios in which multiple devices are arranged in a definite global order. In conventional higher-order quantum theory, such situations are described by the hierarchy of quantum combs, where each local operation has a fixed input-output orientation and all laboratories are causally ordered [3]. An appropriate extension of this hierarchy should allow agents to implement operations in either input-output direction within their local laboratories, while still preserving a well-defined global causal order.

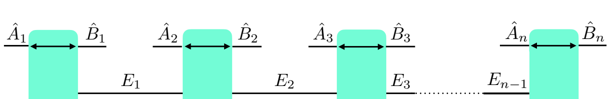

Figure 3: A bi-tooth quantum comb.

This setting can be regarded as a special case of a more general framework in which causally ordered agents implement operations of a given type within their local laboratories. We therefore introduce the notion of a *deterministic network of higher-order maps* and prove that it always defines an admissible higher-order transformation. For n higher-order maps of types $\{x_i\}_{i=1}^n$, the network is represented by the Choi operator $M_1 * \cdots * M_n$, corresponding to the sequential composition of higher-order maps with Choi operators $M_i \in \mathsf{T}_1(\overline{x_i} \to (E_{i-1} \to E_i))$, where $\overline{x_i} = x_i \to I$ (with I the trivial system), so that $\overline{x_i}$ are functionals on x_i. The distinctive feature of this construction is that its local building blocks (higher-order maps) may themselves represent processes with indefinite causal structure. The resulting framework therefore captures scenarios in which causal structure can be locally indefinite within each laboratory, while remaining globally definite across laboratories, which are constrained to follow a fixed sequential causal order (see Fig. 2).

A particularly relevant subclass arises when each x_i corresponds to a bistochastic channel. The resulting structures, which we call *bi-tooth quantum combs*, can be realized as quantum circuits with open slots through sequential composition of partially bistochastic channels (see Fig. 3). This naturally leads to a further generalization: higher-order transformations acting on bi-tooth combs and producing a quantum channel as output. We call these transformations *bi-slot quantum combs*. They correspond to networks of higher-order maps in which each x_i is a functional on bistochastic channels. Unlike standard quantum combs, bi-slot combs do not admit a definite input-output direction for each individual slot. Nevertheless, as deterministic networks of higher-order maps, they preserve a well-defined global causal order and admit a sequential realization analogous to ordinary quantum networks, now with bistochastic supermaps replacing physical channels (see Fig. 4).

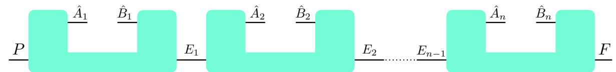

Figure 4: A bi-slot quantum comb.

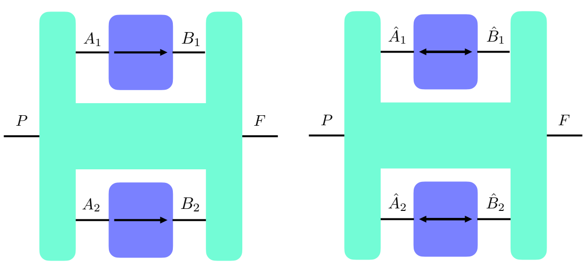

Figure 5: Standard and bistochastic process matrices.

Higher-order transformations without a predefined global causal order are captured by the notion of process matrices [7]. While they enable indefinite causal order and can generate correlations incompatible with any fixed (or probabilistic) causal structure, standard process matrices assume that each party performs a quantum channel. We generalize this notion by allowing each party's intervention to be a higher-order map of type x_i, leading to admissible transformations of type $\mathtt{HPM}_n := \left(\bigotimes_{i=1}^n x_i\right) \to (P \to F)$, which we call *generalized process matrices*. In particular, when the local operations are bistochastic channels, we obtain *bistochastic process matrices* (see Fig. 5), corresponding to transformations of type $\mathtt{BSP}_n := \left(\bigotimes_{i=1}^n (\hat{A}_i \to \hat{B}_i)\right) \to (P \to F)$. Bistochastic process matrices form a strictly richer class than ordinary process matrices: they not only allow superpositions of causal orders but also exploit both input-output directions of the local operations, in particular, generalizing the quantum time flip and the quantum $\mathtt{SWITCH}$ [4]. This richer structure enables stronger correlations (up to the algebraic maximum) such as those realized by the Liu–Chiribella process, which permits perfect two-way signaling [6].

4 Conclusions

In this contribution, we developed a higher-order quantum framework based on bistochastic transformations (maps that preserve the maximally mixed state), allowing elementary devices to be used in both temporal directions without a predefined input-output assignment. Through an operational notion of admissibility, we obtained a recursive characterization of higher-order maps involving bidirectional devices and introduced bistochastic extensions of quantum combs and process matrices, including bi-slot quantum combs, which preserve global causal order while relaxing local input-output orientation. This framework captures processes such as the quantum time flip, its combination with the quantum $\mathtt{SWITCH}$, and the Liu–Chiribella process, which lie beyond standard higher-order quantum theory. Beyond its structural contributions, the framework opens new directions for exploring bidirectional operations as resources in communication, computation, channel discrimination, thermodynamics, and quantum causal modeling, where allowing devices without fixed input-output roles may fundamentally reshape the relation between compositional structure and signaling.

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